

Please explain the following in your own words.

Q1. What is the difference between a map and a flat map?

A map is one-to-one function and a flat map is one-to-many function

Q2. What is the difference between RDD and DataFrame?

RDD stands for Resilient Distributed Dataset, which is fault tolerant and always in partition form [p1, p2, p3]

Whereas data frames are distributed collections of data organized into named columns

Q3. How to control parallelism in Spark.

In Apache Spark, parallelism is determined by the total number of partitions across your dataset and the number of executor cores available to process them simultaneously. Controlling parallelism effectively ensures optimal resource utilization without causing memory pressure or severe data skew.

Q4. Please explain the following

- Spark Driver

It is a program where we create Spark Context. In other words, the spark driver is a driver where the main program is written, which basically has methods, and those methods are always called in the main program. It is also called the entry point of application, where execution is performed.

- Execution Mode

Execution mode refers to how and where your application runs on a cluster specifically dictating where the driver process

- Spark executor

When we launch the application, Spark Context will break it into multiple tasks, and SC submits it to the worker node, and Spark Executor is the service that executes these tasks.

- Task

When application is run in a Spark context, the Spark context breaks it into multiple tasks.

- Stages

Stages is nothing but collection of multiple tasks. In other words it is group of common tasks.

- Worker Node

Worker Node is a term in Spark where it receives the multiple tasks from spark context. The tasks are then executed by the executor and the executor shares the result to the driver program or Spark context.